Machine Learning

K-Means (Clustering Algorithm)

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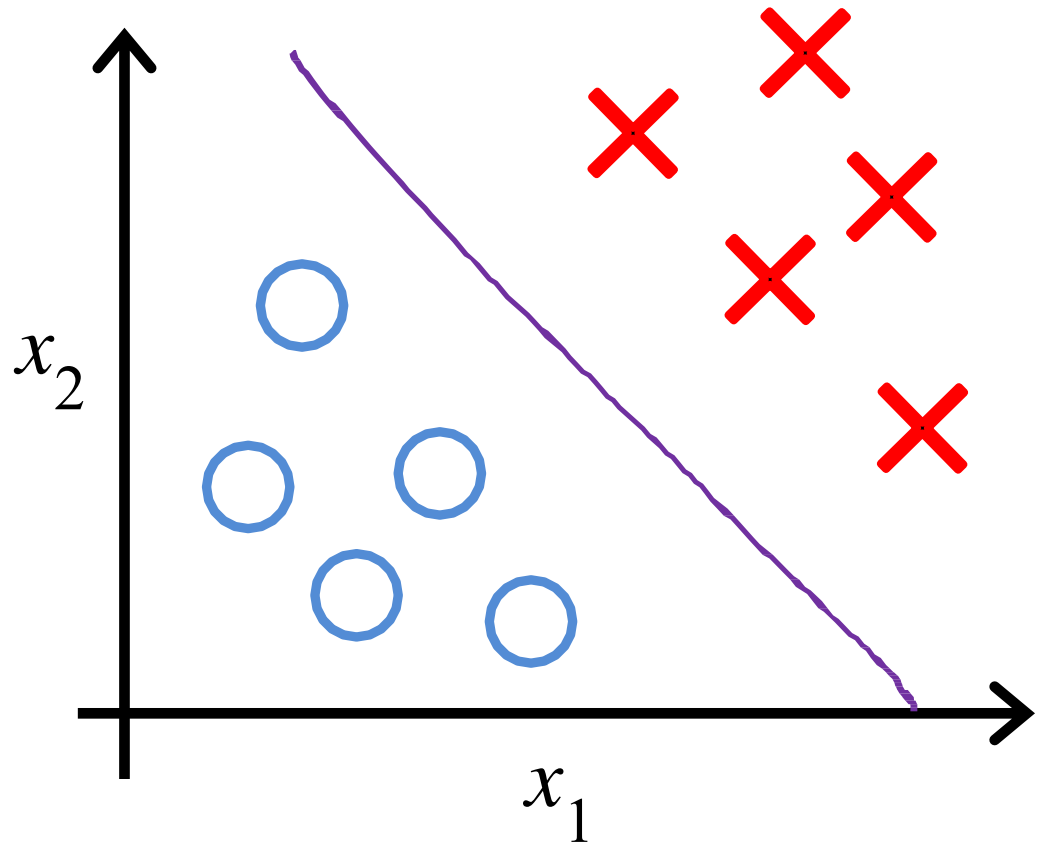
Outline

- What is clustering
- Applications of Clustering Algorithm Example
- What is K-Mean
- K-Mean Algorithm
- Example of K-Mean
- Optimization Objective
- Initializing K-Mean
- Choosing the number of clusters
- Programming Assignement



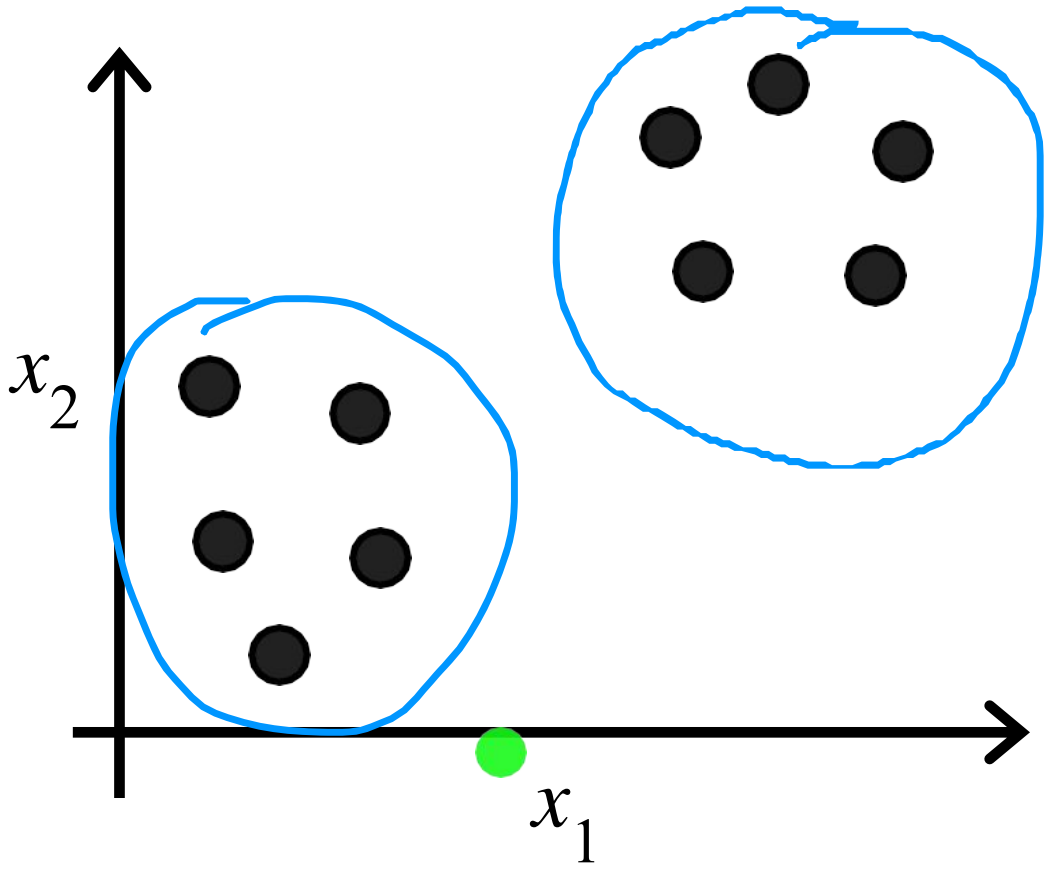
What is clustering?

Supervised learning



Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \ldots, (x^{(m)}, y^{(m)})\}$?

Unsupervised learning



Clustering

Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(m)}\}$



Applications of Clustering Algorithm

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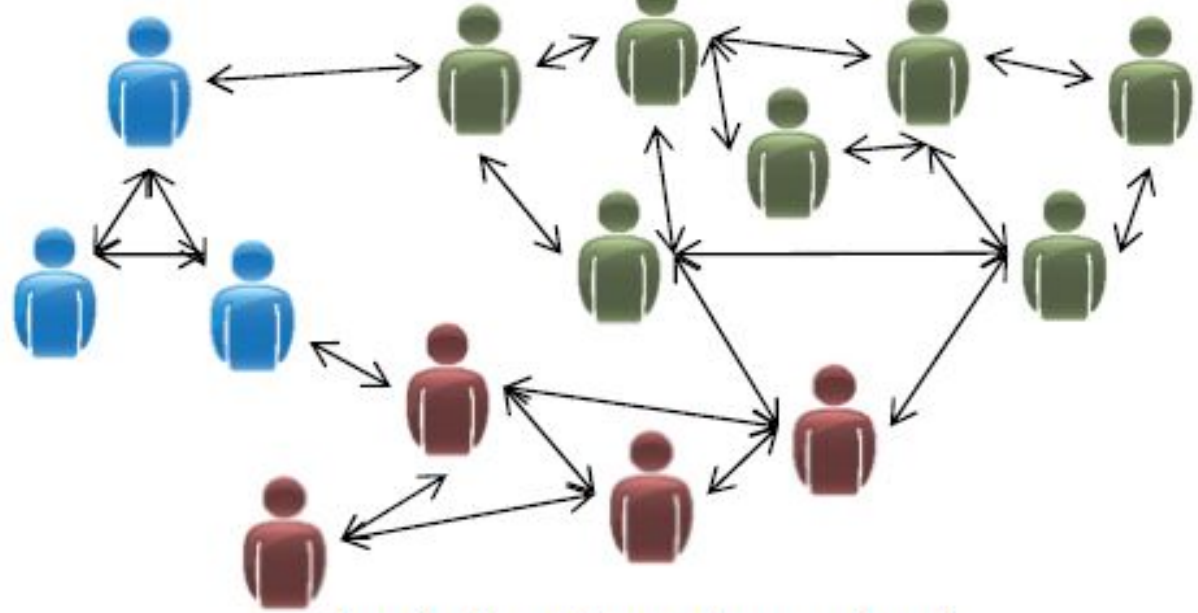


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Organize computing clusters



Social network analysis



Market segmentation



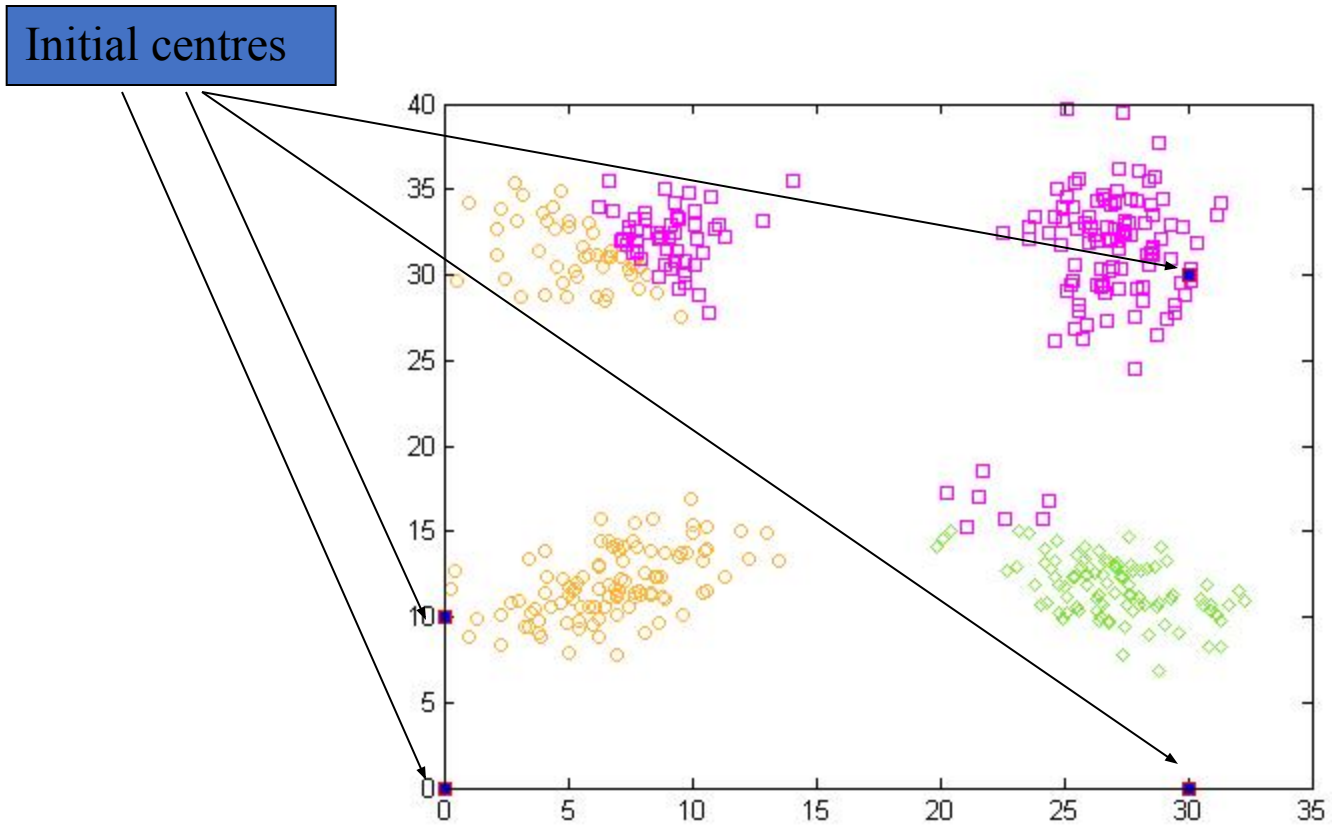
Astronomical data analysis



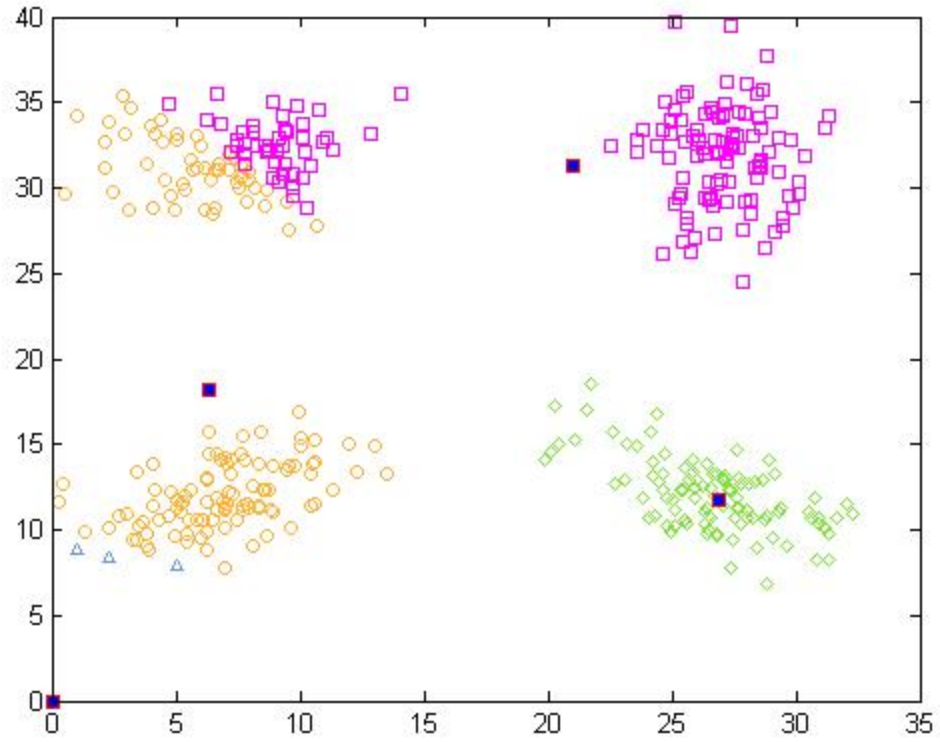
Q & A



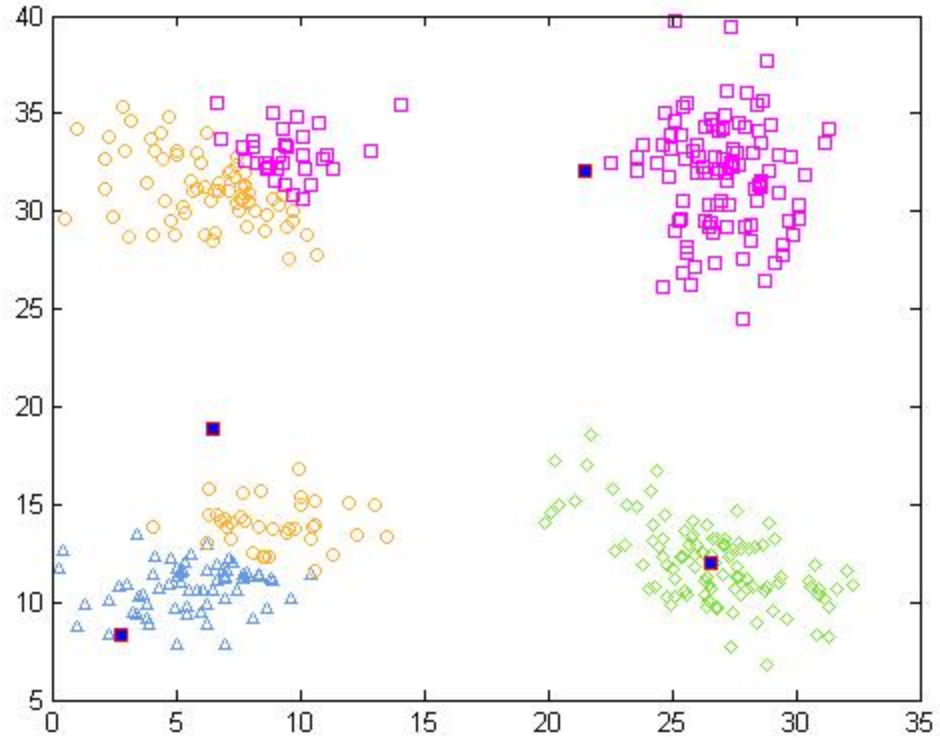
K-Mean (Clustering Algorithm)



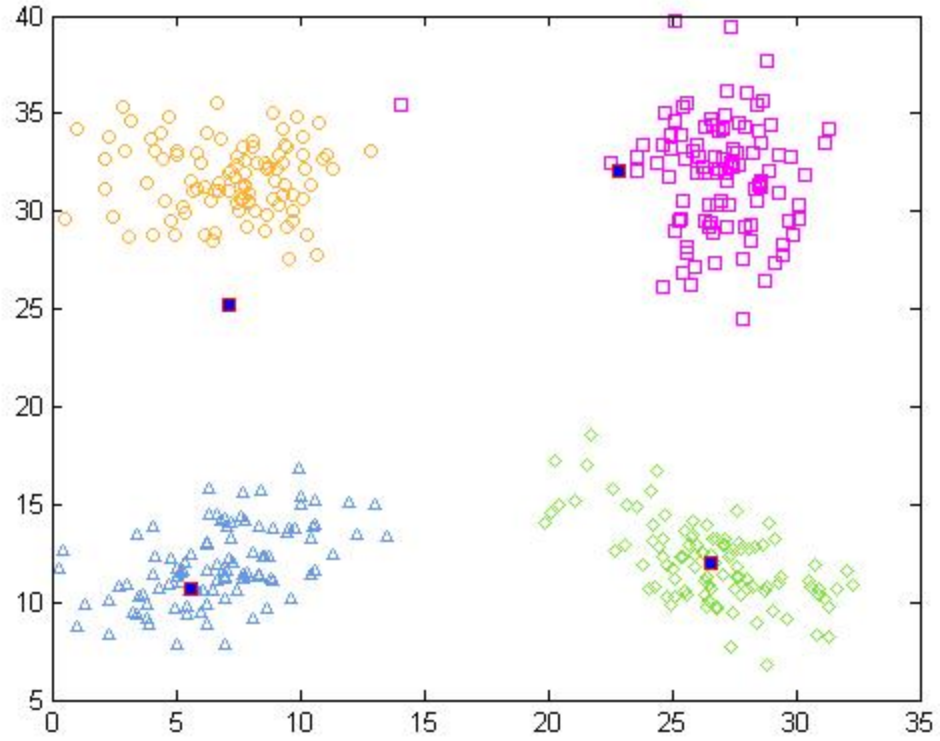
Ref: K. Javed et. al, “The behavior of K-Means: An empirical study”, International Conference on Electrical Engineering, 2008



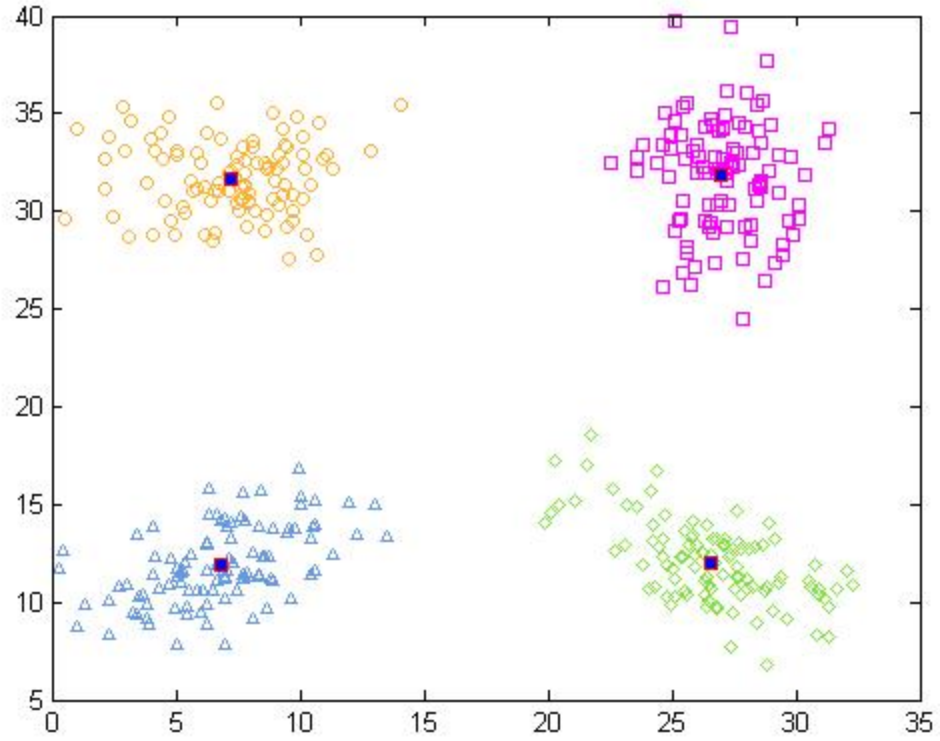
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K-Mean Algorithm

K-Means Algorithm

- 1.Initialize:
  - Choose K random data points from the data set to represent the initial centers of the K partitions

- 2.Calculate the group memberships:

$$b_i^t = \begin{cases} 1 \text{ if } \left\|\mathbf{x}^t - \mathbf{m}_i\right\| = \min_j \left\|\mathbf{x}^t - \mathbf{m}_j\right\| \\ 0 \text{ otherwise} \end{cases}$$

- 3.Update the centroids:

$$\mathbf{m}_i = \frac{\sum_t b_i^t \mathbf{x}^t}{\sum_t b_i^t}$$

- 4.Repeat steps 2 and 3
  - until converge to stable values



Q & A



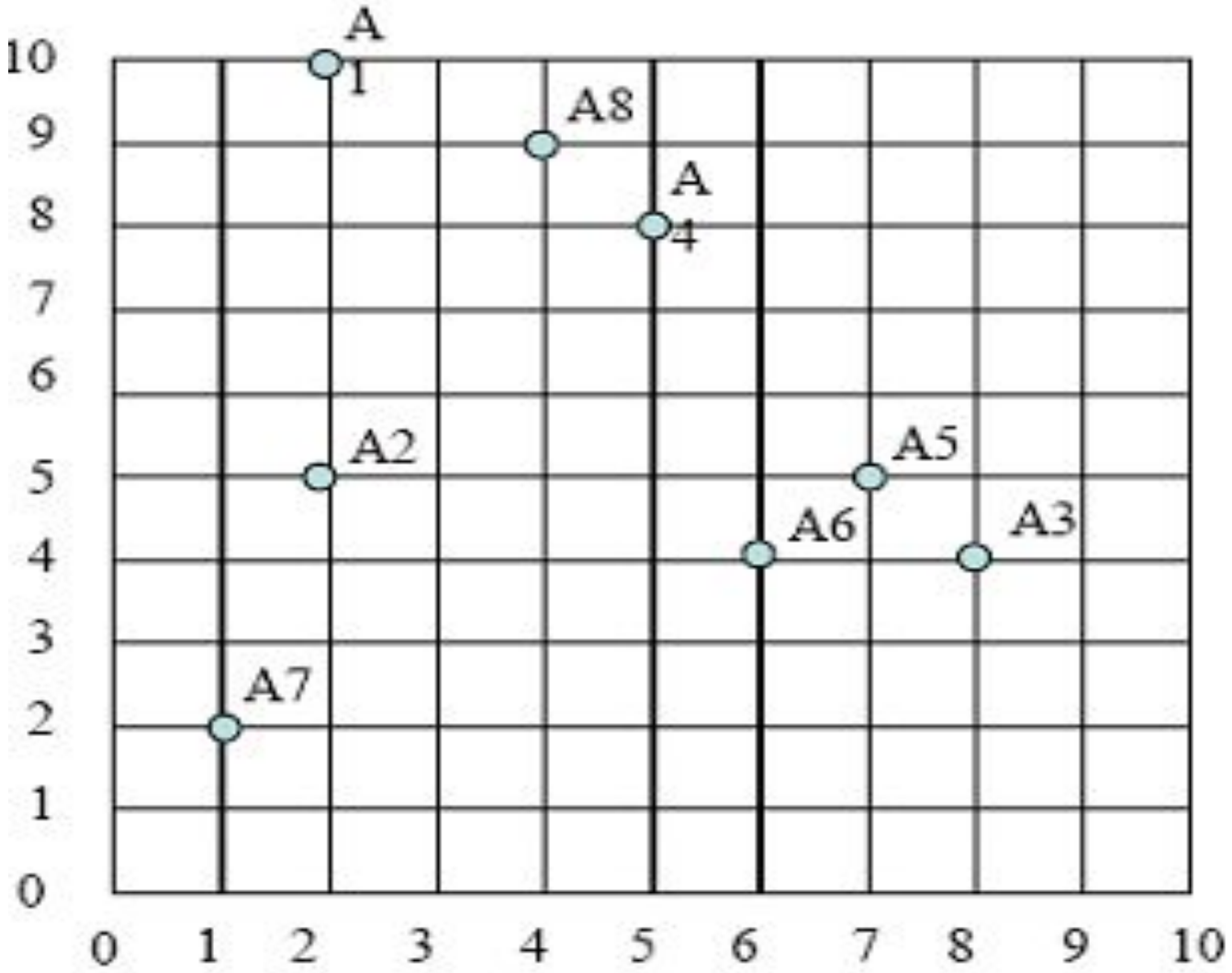
Example of K-Mean Algorithm

Example

- Problem: Cluster the following eight points (with (x1, x2) representing locations) into three clusters
  - A1(2, 10)
  - A2(2, 5)
  - A3(8, 4)
  - A4(5, 8)
  - A5(7, 5)
  - A6(6, 4)
  - A7(1, 2)
  - A8(4, 9).
- Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).
- The distance function between two points $a=(x1, y1)$ and $b=(x2, y2)$ is defined as: $\rho(a, b) = |x2 - x1| + |y2 - y1|$.

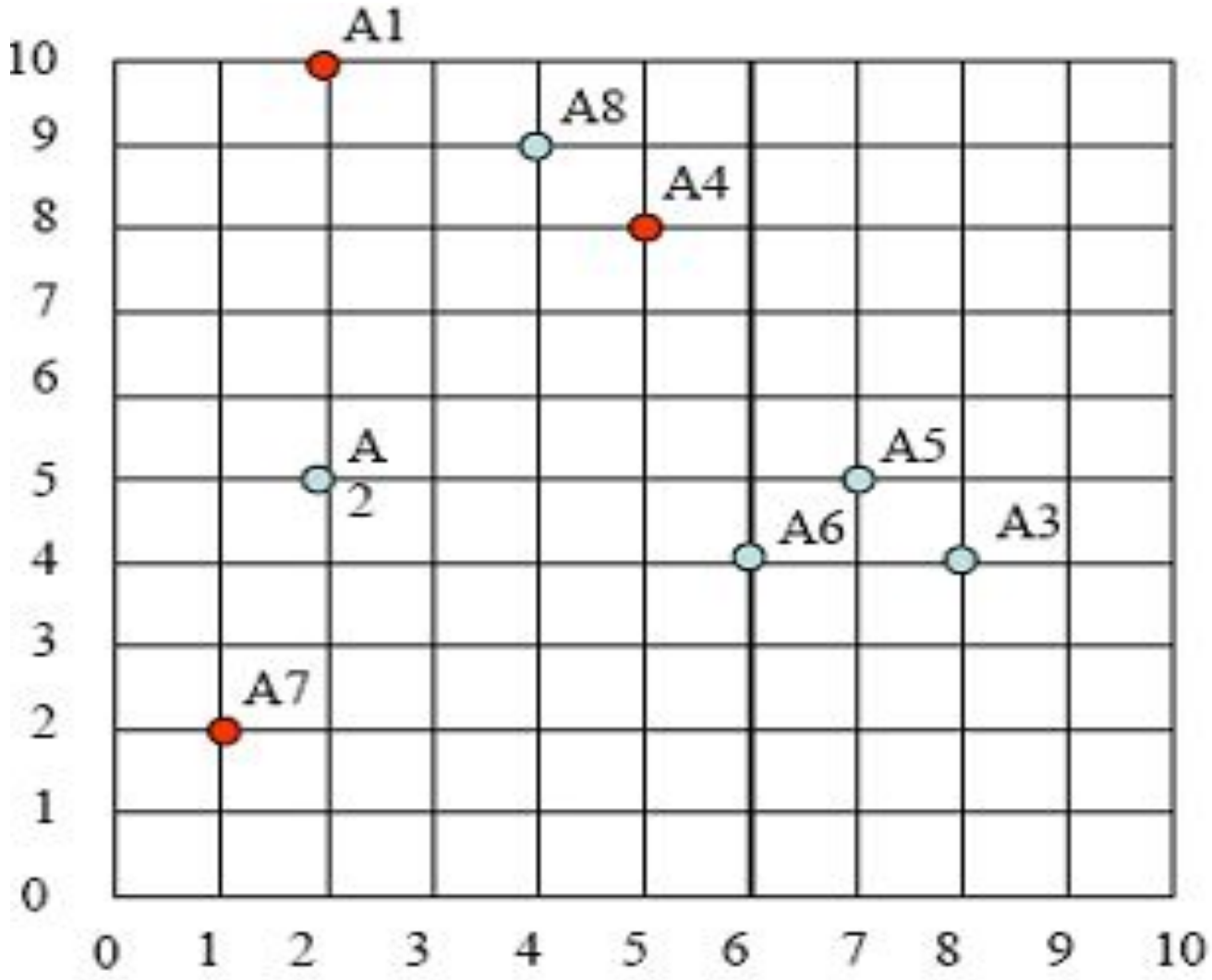
Plot

A1(2, 10)
A2(2, 5)
A3(8, 4)
A4(5, 8)
A5(7, 5)
A6(6, 4)
A7(1, 2)
A8(4, 9).



Plot

A1(2, 10)
A2(2, 5)
A3(8, 4)
A4(5, 8)
A5(7, 5)
A6(6, 4)
A7(1, 2)
A8(4, 9).



Solution (Iteration 1)

		(2, 10)	(5, 8)	(1, 2)	
	Point	**Dist Mean 1**	**Dist Mean 2**	**Dist Mean 3**	**Cluster**
A1	(2, 10)				
A2	(2, 5)				
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

$$|x2 - x1| + |y2 - y1|$$

Solution (Iteration 1)

		(2, 10)	(5, 8)	(1, 2)	
	Point	**Dist Mean 1**	**Dist Mean 2**	**Dist Mean 3**	**Cluster**
A1	(2, 10)	0	5	9	1
A2	(2, 5)				
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

$$|x2 - x1| + |y2 - y1|$$

Solution (Iteration 1)

		(2, 10)	(5, 8)	(1, 2)	
	Point	**Dist Mean 1**	**Dist Mean 2**	**Dist Mean 3**	**Cluster**
A1	(2, 10)	0	5	9	1
A2	(2, 5)	5	6	4	3
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

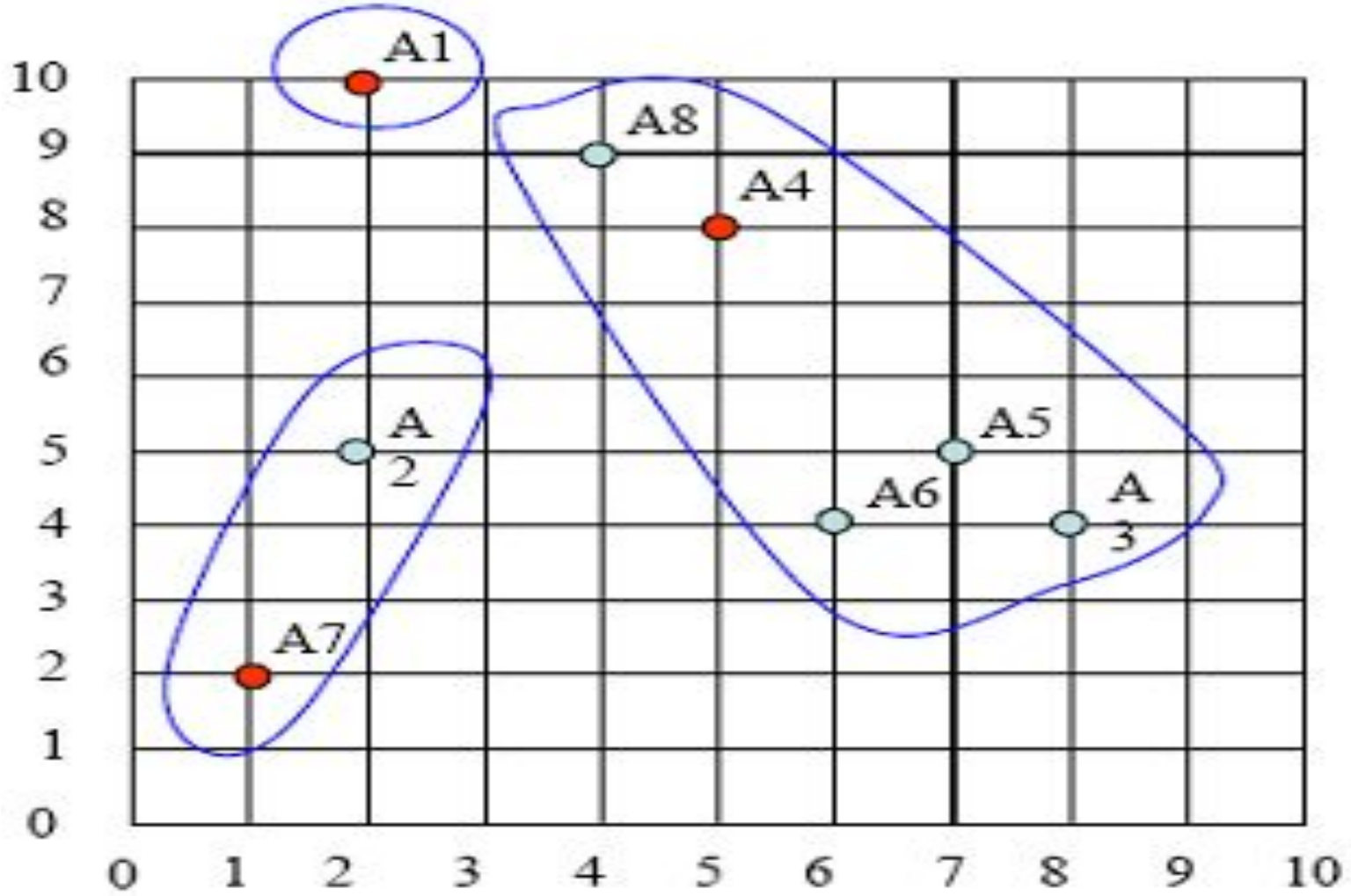
$$|x2 - x1| + |y2 - y1|$$

Solution (Iteration 1)

		(2, 10)	(5, 8)	(1, 2)	
	Point	**Dist Mean 1**	**Dist Mean 2**	**Dist Mean 3**	**Cluster**
A1	(2, 10)	0	5	9	1
A2	(2, 5)	5	6	4	3
A3	(8, 4)	12	7	9	2
A4	(5, 8)	5	0	10	2
A5	(7, 5)	10	5	9	2
A6	(6, 4)	10	5	7	2
A7	(1, 2)	9	10	0	3
A8	(4, 9)	3	2	10	2

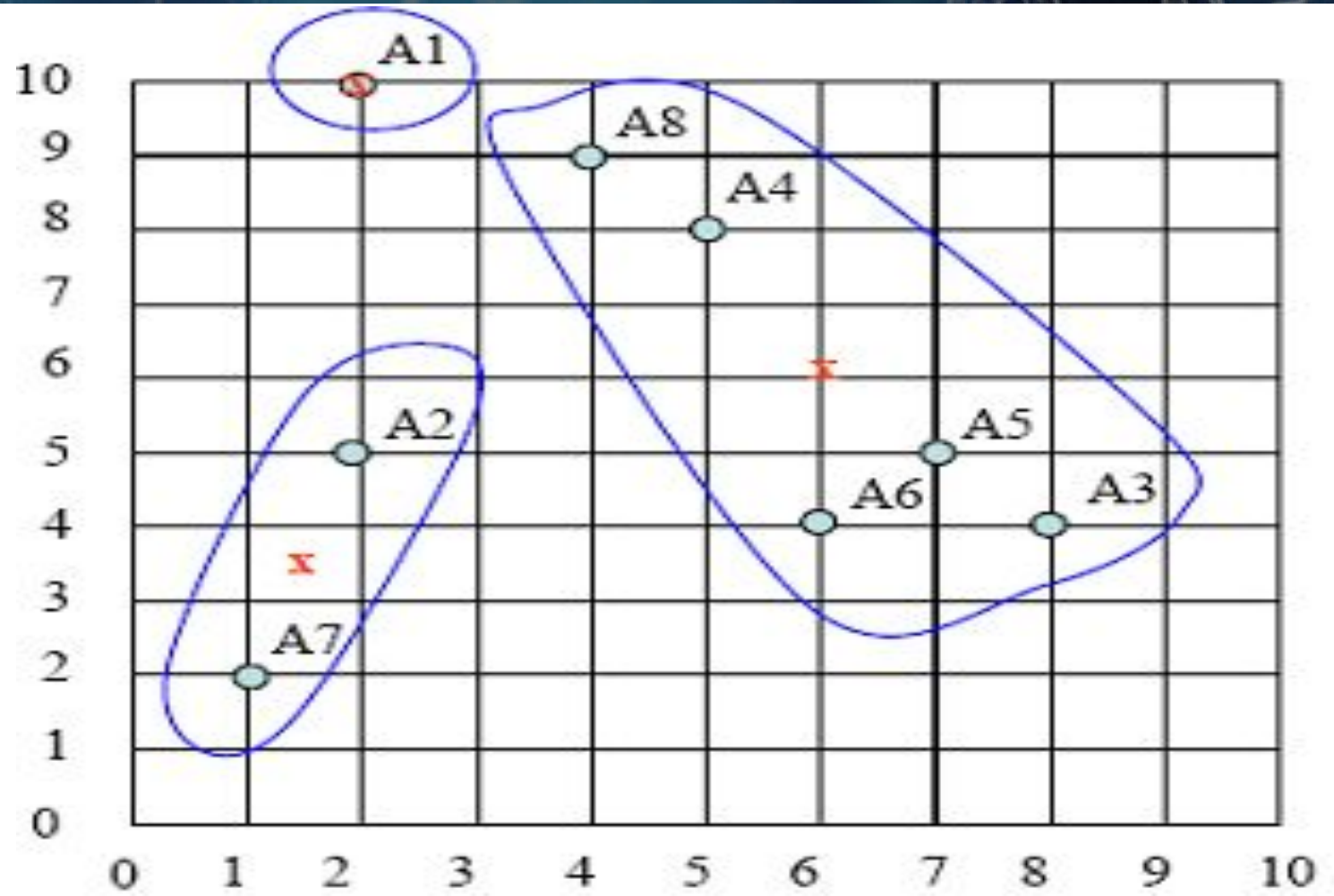
$$|x2 - x1| + |y2 - y1|$$

Membership



	Point	**Cluster**
A1	(2, 10)	1
A2	(2, 5)	3
A3	(8, 4)	2
A4	(5, 8)	2
A5	(7, 5)	2
A6	(6, 4)	2
A7	(1, 2)	3
A8	(4, 9)	2

Centroid Update



	Point	**Cluster**
A1	(2, 10)	1
A2	(2, 5)	3
A3	(8, 4)	2
A4	(5, 8)	2
A5	(7, 5)	2
A6	(6, 4)	2
A7	(1, 2)	3
A8	(4, 9)	2

Solution (Iteration 2)

		(2, 10)	(6, 6)	(1.5, 3.5)	
	Point	**Dist Mean 1**	**Dist Mean 2**	**Dist Mean 3**	**Cluster**
A1	(2, 10)				
A2	(2, 5)				
A3	(8, 4)				
A4	(5, 8)				
A5	(7, 5)				
A6	(6, 4)				
A7	(1, 2)				
A8	(4, 9)				

$$|x2 - x1| + |y2 - y1|$$

d)

We would need two more epochs. After the 2nd epoch the results would be:

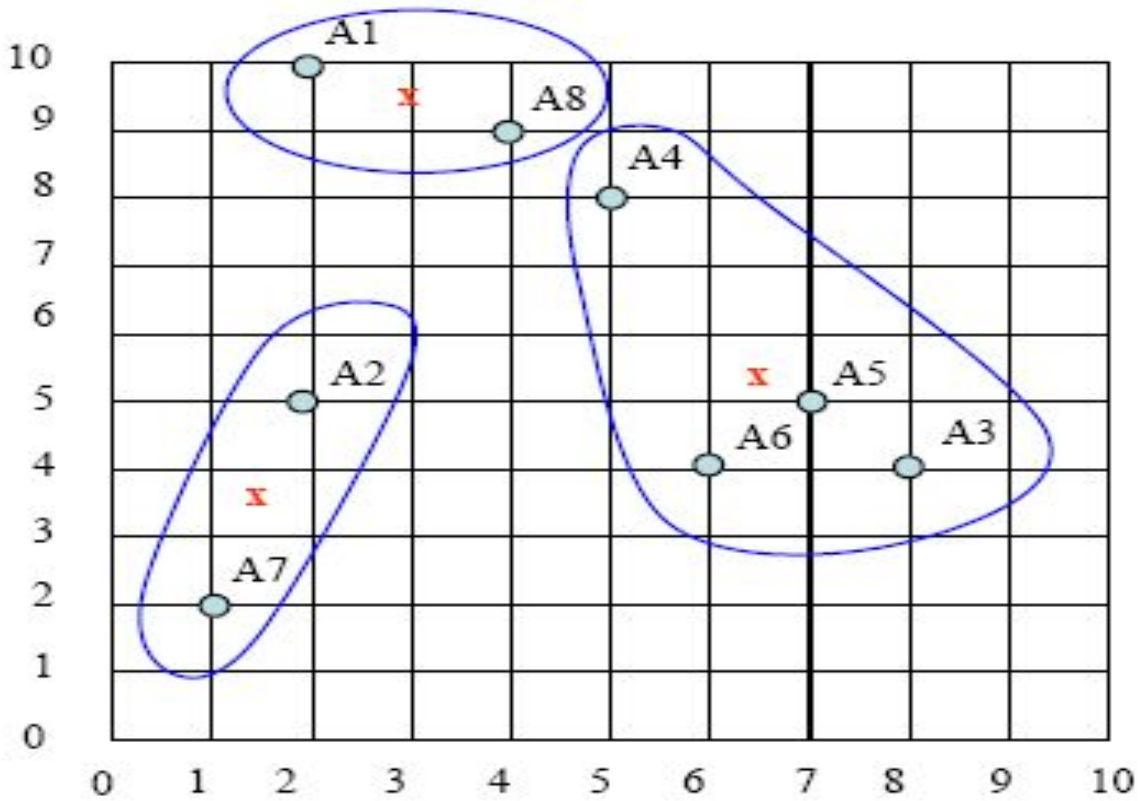
1: {A1, A8}, 2: {A3, A4, A5, A6}, 3: {A2, A7}

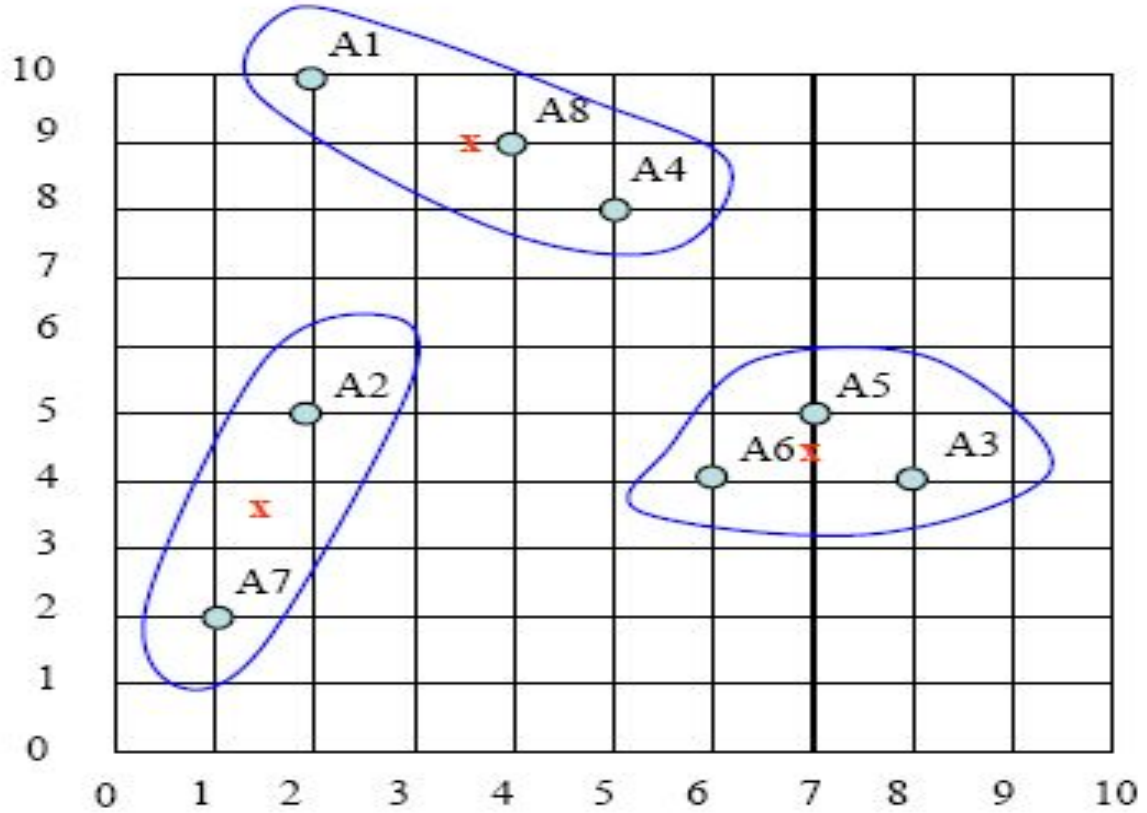
with centers C1=(3, 9.5), C2=(6.5, 5.25) and C3=(1.5, 3.5).

After the 3rd epoch, the results would be:

1: {A1, A4, A8}, 2: {A3, A5, A6}, 3: {A2, A7}

with centers C1=(3.66, 9), C2=(7, 4.33) and C3=(1.5, 3.5).







Q & A



Optimization objective

K-means optimization objective

$c^{(i)}$ = index of cluster $(1, 2, \ldots, K)$ to which example $x^{(i)}$ is currently assigned

μ_k = cluster centroid k

$\mu_{c^{(i)}}$ = cluster centroid of cluster to which example $x_{(i)}$ has been assigned

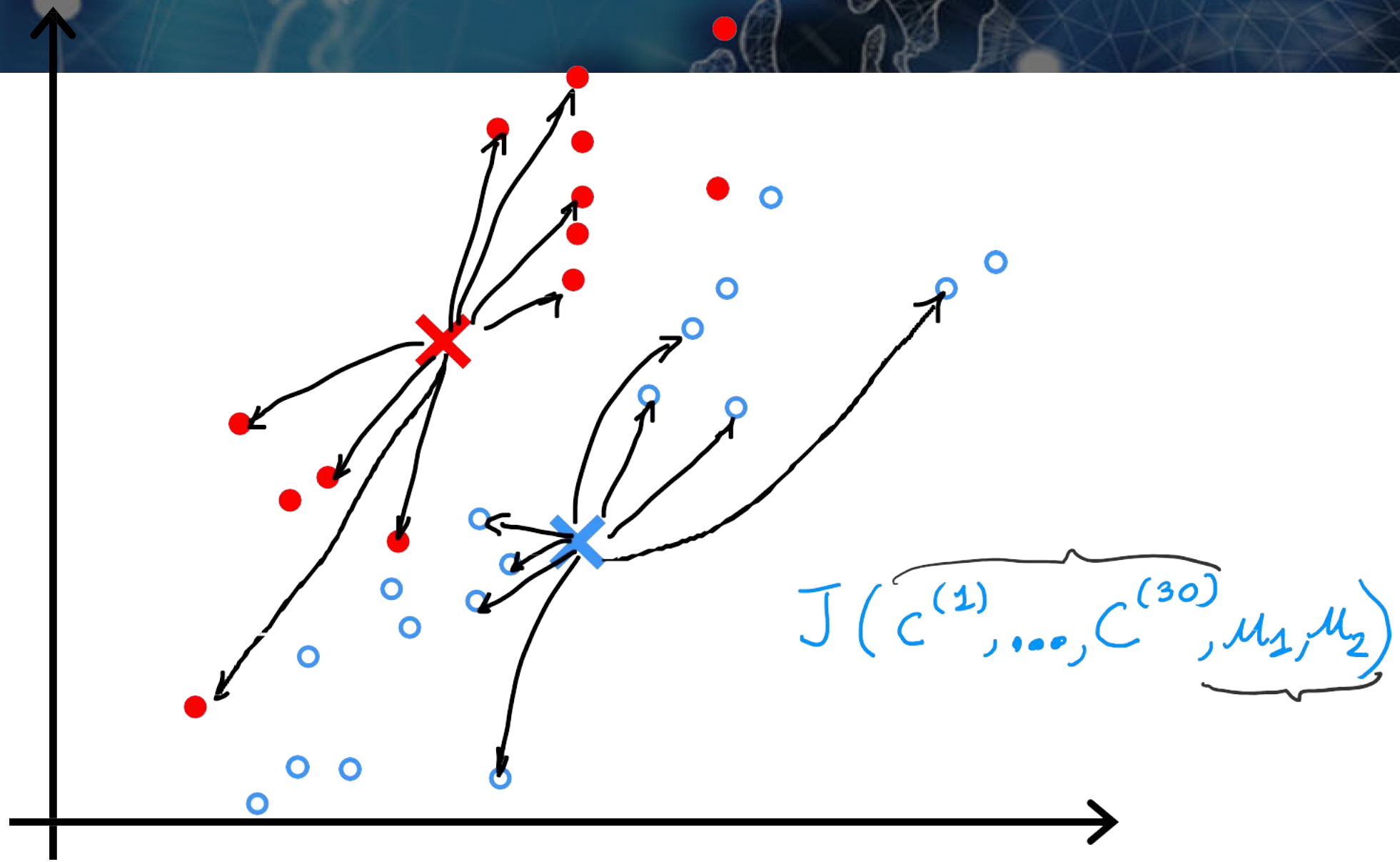
$x^{(10)}$ $c^{(10)}$ $\mu_c^{(10)}$

Cost function

$$J(c^{(1)}, \ldots, c^{(m)}, \mu_1, .., \mu_K) = \frac{1}{m} \sum_{i=1}^{m} \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

$$\min_{c^{(1)}, \ldots, c^{(m)}} J(c^{(1)}, \ldots, c^{(m)}, \mu_1, .., \mu_K)$$

$$\mu_1, \ldots, \mu_K$$



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Cost function for K-means

$$J\left(c^{(1)}, \ldots, c^{(m)}, \mu_1, \ldots, \mu_K\right) = \frac{1}{m}\sum_{i=1}^{m}\left\|x^{(i)} - \mu_{c^{(i)}}\right\|^2$$

Repeat {

Assign points to cluster centroids

for i = 1 to m

$c^{(i)}$:= index of cluster centroid closest to $x^{(i)}$

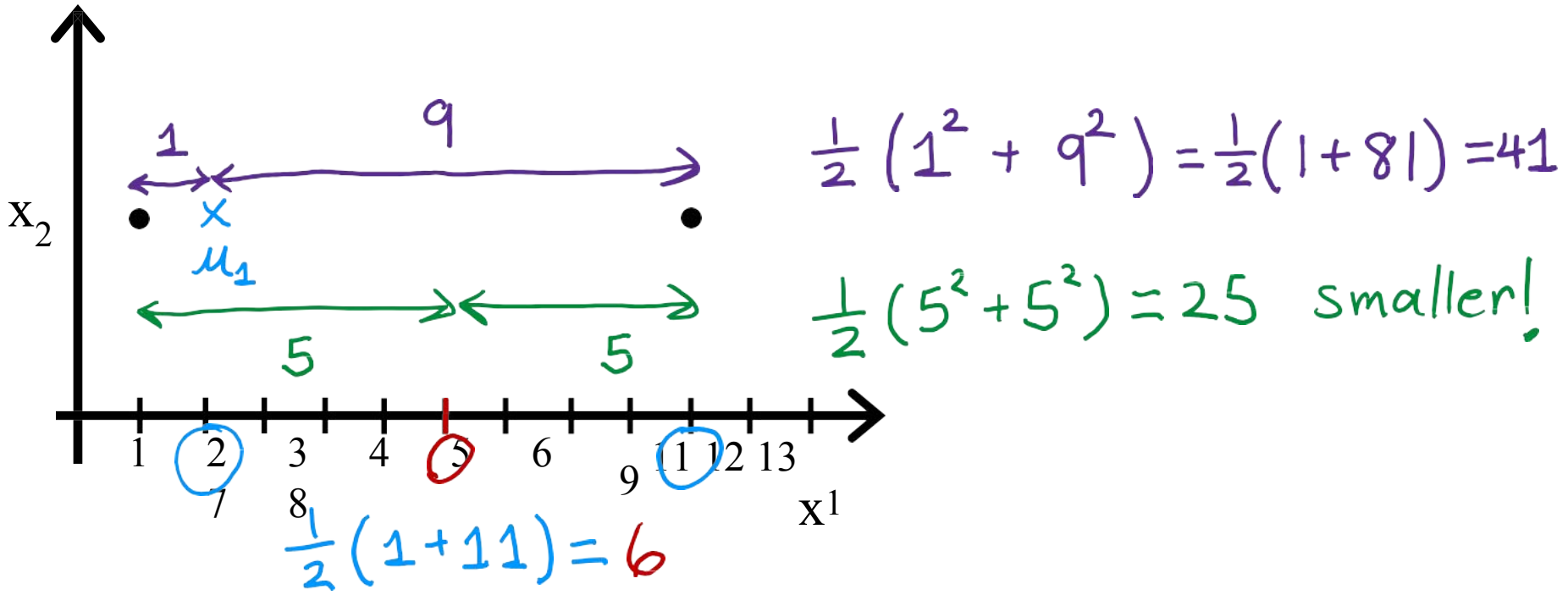
Move cluster centroids

for k = 1 to K

μ_k := average of points in cluster k

}

Moving the centroid





Initializing K-means

K-means algorithm

Step 0: Randomly initialize K cluster centroids $\mu_1, \mu_1, \ldots, \mu_k$

Repeat {

Step 1: Assign points to cluster centroids

Step 2: Move cluster centroids

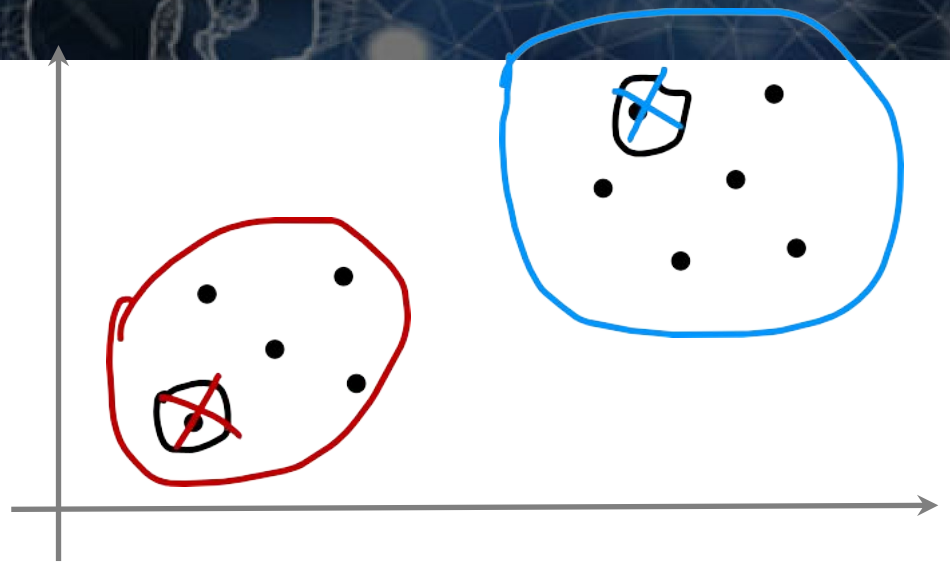
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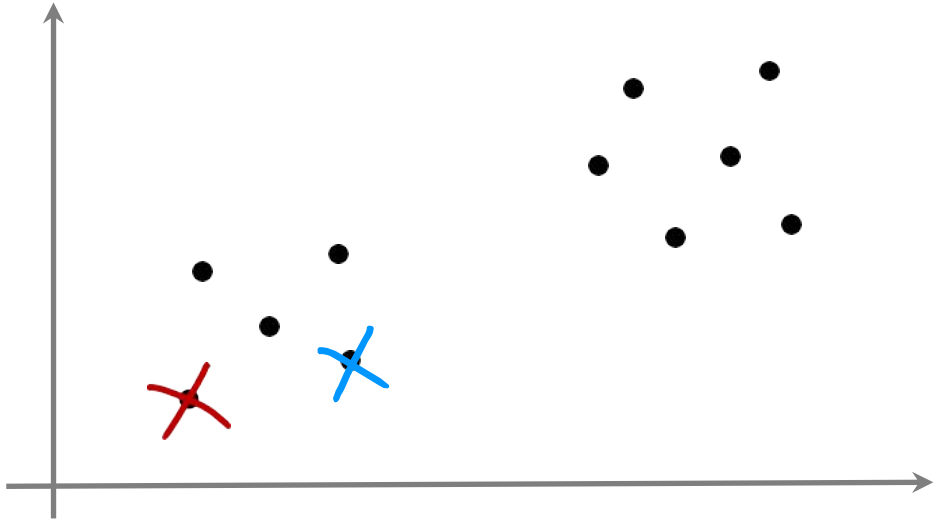
Random initialization

Choose $K < m$

Randomly pick K training examples.

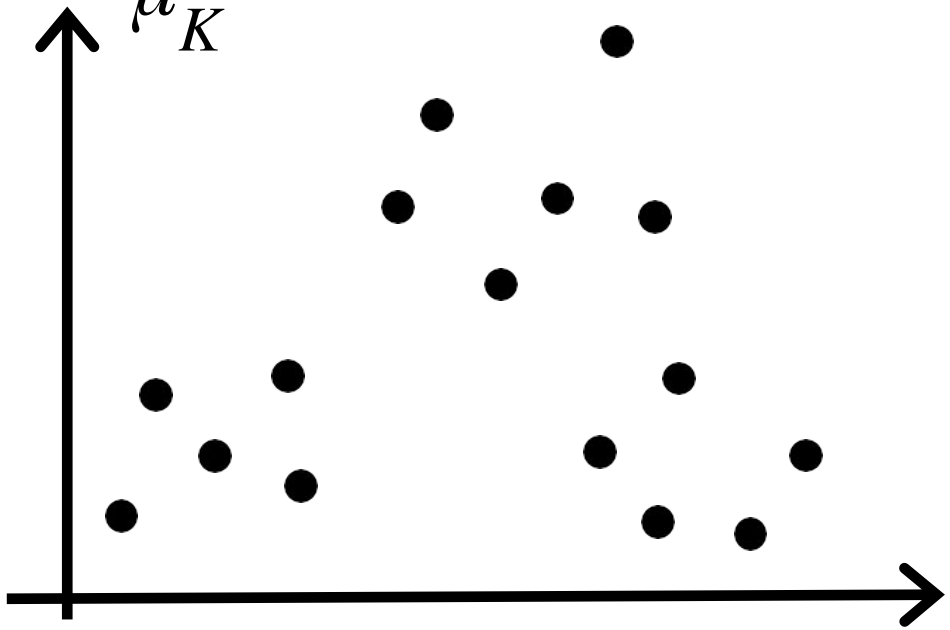
Set $\mu_\$$, $\mu_\$$,..., μ_k equal to these K examples.

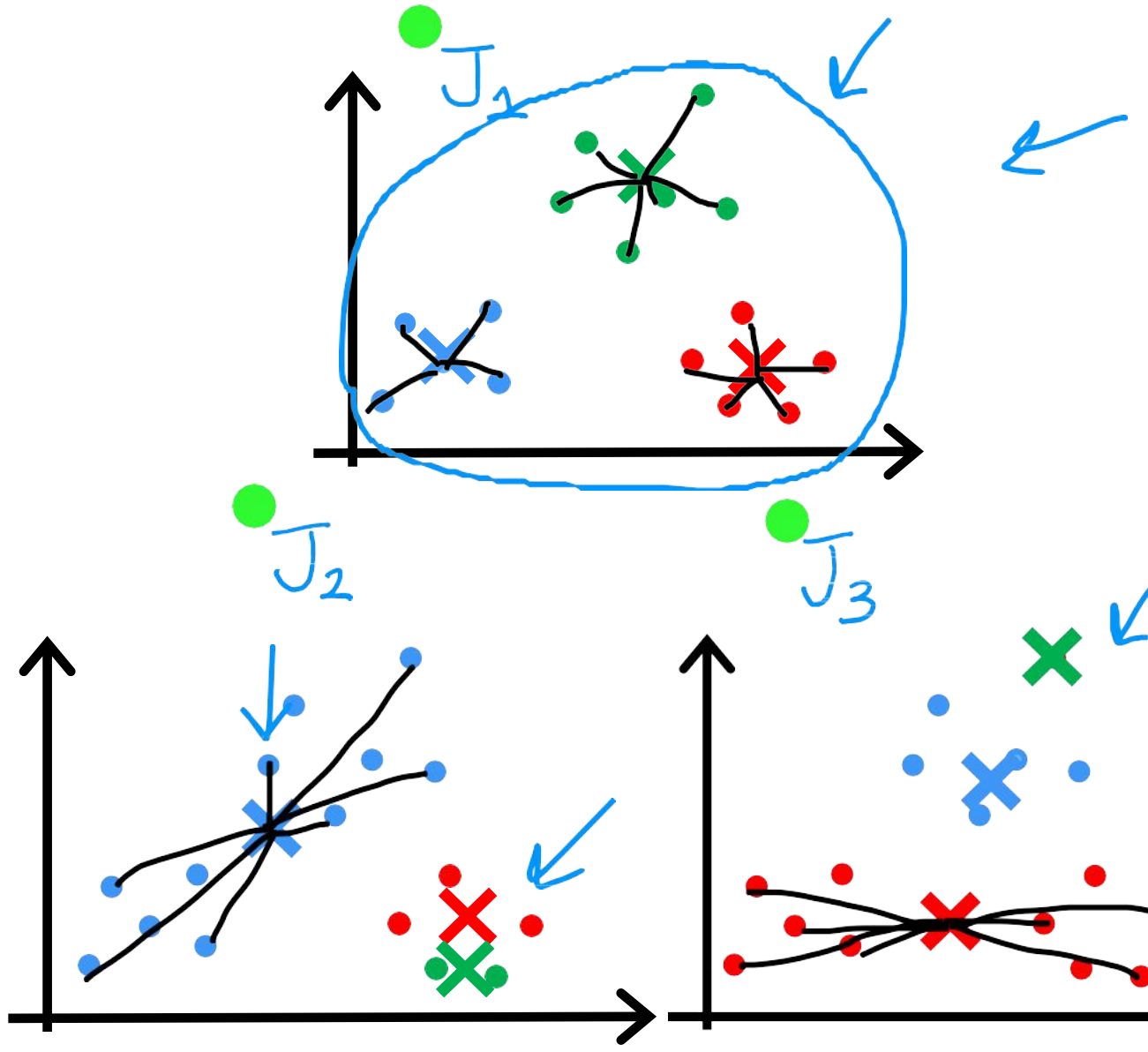




K=3

$$J(c^{(1)}, \ldots, c^{(m)}, \mu_1, \ldots, \mu_K)$$





Random initialization

better than running K-means once

For $i = 1$ to 100 {

50-1000

Randomly initialize K-means.

pick K random examples set as clustercentroids

Run K-means. Get $c^{(1)}, \ldots, c^{(m)}, \mu_1, \mu_1, \ldots, \mu_k$

Compute $J(c^{(1)}, \ldots, c^{(m)}, \mu_1, \mu_1, \ldots, \mu_k)$

distortion (cost function)

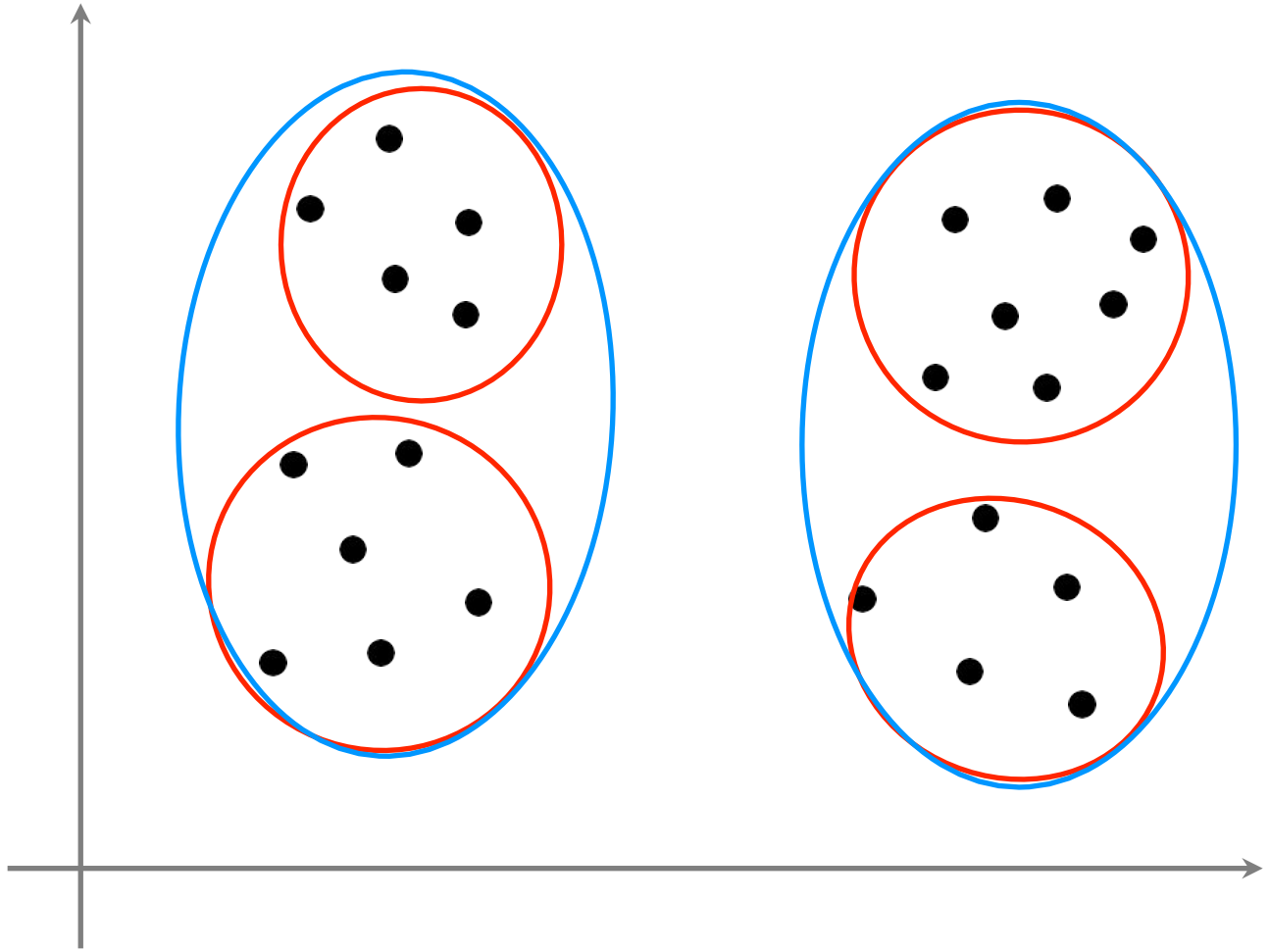
}

Pick set of clusters that gave lowest cost J



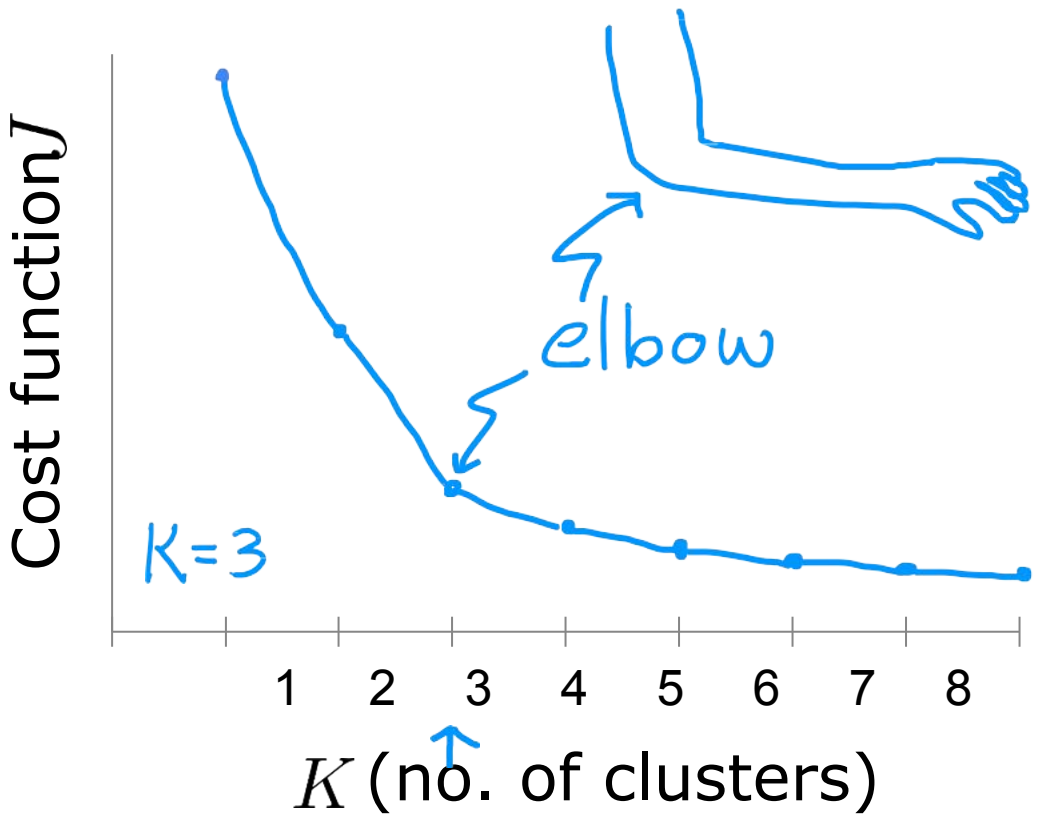
Choosing the Number of Clusters

What is the right value of K?



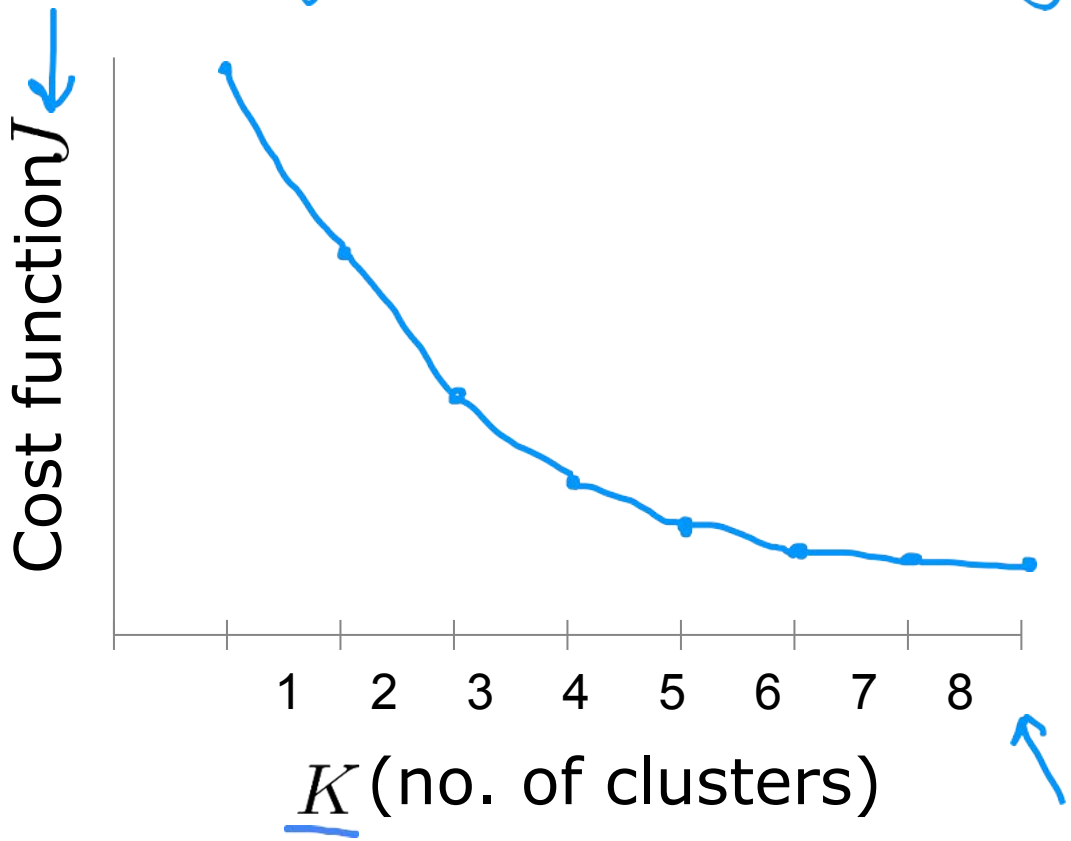
Choosing the value of K

Elbow method



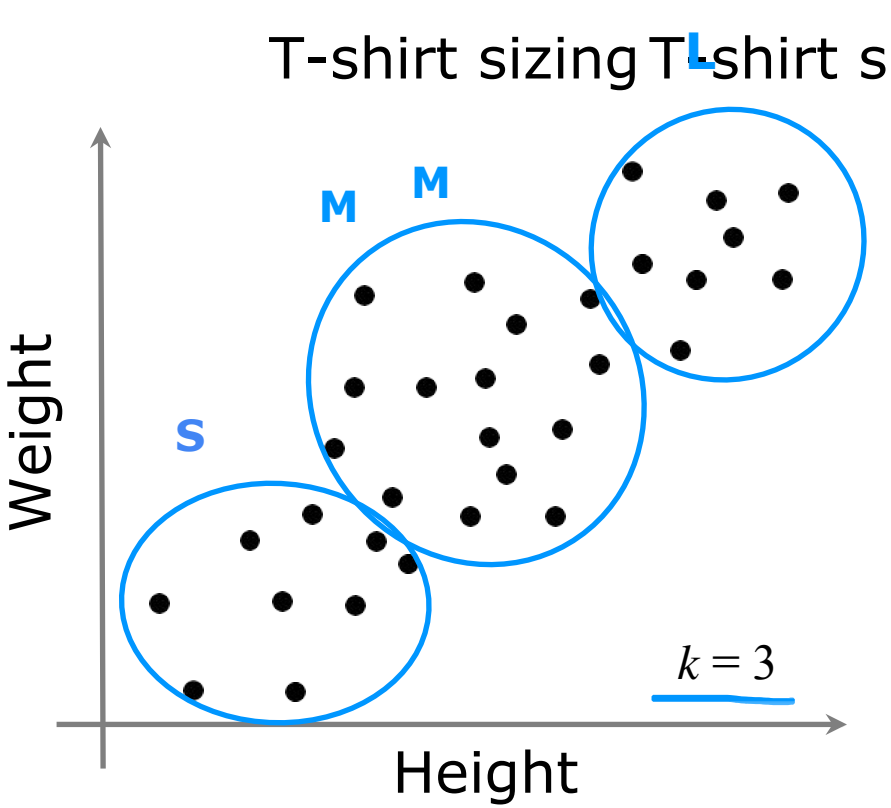
the right "K" is often ambiguous

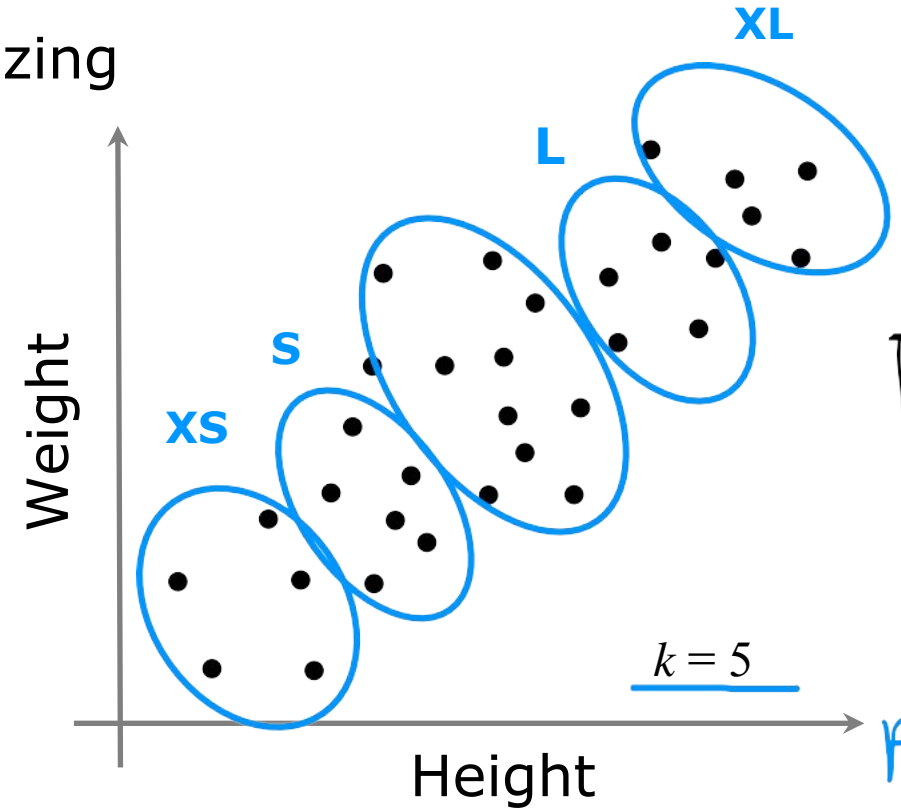
Don't choose K just to minimize cost J

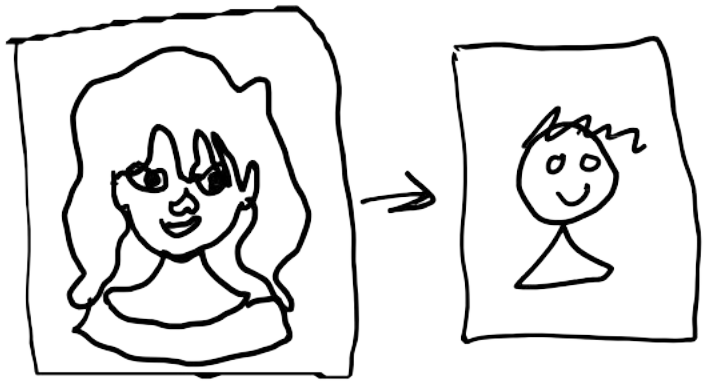


Choosing the value of K

Often, you want to get clusters for some later (downstream) purpose.
Evaluate K-means based on how well it performs on that later purpose.







practice lab:
image compression!



Programming Assignment

k=2 k=3 k=10



Reference: Bishop 2006.





Questions